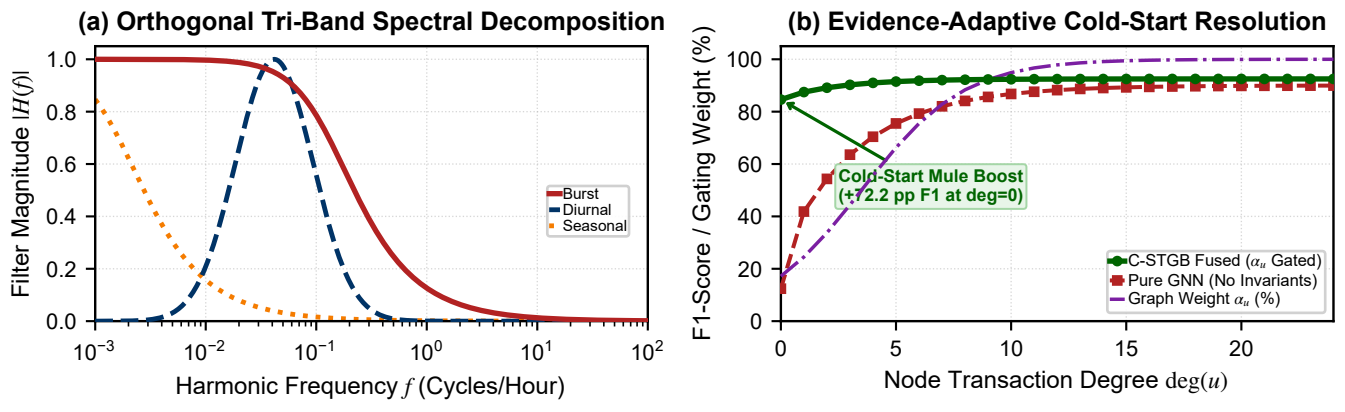

(a) Orthogonal Tri-Band Spectral Decomposition
Filter Magnitude $|H(f)|$
1.0
0.8
0.6
0.4
0.2
0.0
$10^{-3}$
$10^{-2}$
$10^{-1}$
$10^{0}$
$10^{1}$
$10^{2}$
Harmonic Frequency $f$ (Cycles/Hour)
Burst
Diurnal
Seasonal
(b) Evidence-Adaptive Cold-Start Resolution
F1-Score / Gating Weight (%)
100
80
60
40
20
0
0
5
10
15
20
Node Transaction Degree $\deg(u)$
Cold-Start Mule Boost
(+72.2 pp F1 at deg=0)
C-STGB Fused ($\alpha_u$ Gated)
Pure GNN (No Invariants)
Graph Weight $\alpha_u$ (%)